

H15 Semantic information through natural scrambles: which information keeps agents and the swarm viable

not yet rated

level: hypothesis · pipeline checks 85% · updated —

exploratory round 1 done (2026-10-03, non-holdout only). The pre-registered "memory is load-bearing" predictions failed: no detectable day-scale viability cost of losing up to ~80% of memory (P1) or of starting with an empty memory (P4); the rewrite control...

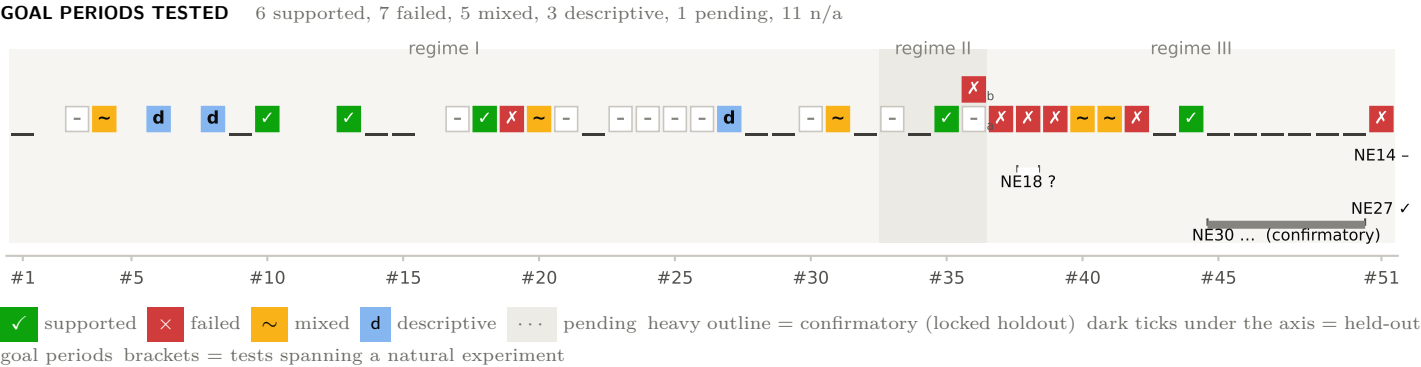
WHAT HAPPENED

Not yet summarized. The question: Kolchinsky–Wolpert: information is semantic, or load-bearing, if scrambling it lowers the system’s viability. We can’t run new swarms or replay agents, but the logs contain natural scrambles: memory loss and resets, a change in consolidation rate (NE14), newcomers with empty memories, room cuts that remove chat exposure, and widened history search (NE18). Which removals lower viability, by how much, and what fraction of the available information is load-bearing (η)? Practical payoff: which context actually keeps a swarm working.

PHYSICAL MODEL

From: physics-models/04-semantic-information (ΔV under scrambling, η = load-bearing fraction), with H01’s architecture.md (viability candidates, D2.6 homeostasis rule for choosing V) and notes/kolchinsky-qa-transcript.md. ### Mapping onto Kolchinsky–Wolpert (KW) - System X: one agent. Its state has three parts with different lifetimes: - long-term memory M (agent_memories; summarized without text in memory_stats), ca

Pending.



OBSERVABLES

Figure pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Bottom line. Pending.

WHAT THE FIGURE SHOWS

Pending.